
Data Preprocessing (EDA & Cleaning)

1. Dataset Overview

The dataset predicts whether a person earns more than \$50K per year. It has 15 features including age, education, job type, marital status, race, sex, work hours, and country of origin. The data came in two files: Train.csv and Test.csv, which were merged into one DataFrame for consistent processing. The total dataset had around 32,500 rows.

2. Exploratory Data Analysis

Before cleaning, the data was explored to understand its structure and the relationships between features.

Class Balance: About 75% of people earn \$50K or less and 25% earn more, so the dataset is imbalanced.

Numeric Features: A correlation heatmap showed that education level has the strongest link with salary, followed by age and work hours. Capital gain and capital loss are mostly zero but are strong indicators when they are not.

Categorical Features: Bar charts showed the percentage of high earners in each category. Married people, executives, and self-employed individuals had the highest rates of earning more than \$50K. Single people, those in service jobs, and females had lower rates.

3. Preprocessing Techniques

The following steps were applied to clean the data before modeling:

- Missing values: The dataset used '?' to mark unknowns. These were replaced with proper null values using regex. Rows with missing work-class or position were dropped. Missing native-country values were filled with the most common value (United States).
- Duplicates: 24 duplicate rows were found and removed.
- Dropped columns: The education column was removed because education-num already contains the same information as a number. The work-fnl column was also removed as it is a census weight that adds noise.

4. Feature Engineering

Three new features were created by combining existing columns to capture patterns that single features miss:

- edu-workHours: education level squared times hours per week. Highly educated people who work more hours tend to earn more.
- edu-age: education level squared times age. An older person with higher education has had more time to benefit from their qualifications.

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- `workHours-age`: work hours times age. Combines experience and effort. Both new features were tested using binning and showed a clear increase in high earners across the bins, confirming they are useful.

5. Categorical Encoding

Machine learning models need numbers, so categorical columns were converted using three methods:

- Binary encoding: The sex column was converted to 1 for Male and 0 for Female.
- One-Hot Encoding: Used for marital-status, relationship, and race. Each category becomes its own column with a 0 or 1, because these columns have no natural order.
- Target Encoding: Used for work-class, position, and native-country. Each category is replaced by the average salary rate of people in that group, calculated only from the training data. A smoothing value of 70 was used to avoid overfitting on rare categories.

6. Data Splitting & Normalization

After encoding, the data was split into three sets: training (21,498 rows), validation (4,598 rows), and test (4,598 rows). The test data was split 50/50 into validation and test to give two separate evaluation sets.

All features were then normalized using Z-score standardization (`StandardScaler`). This rescales every feature to have a mean of zero and a standard deviation of one. It is important because some features have very large values while others are small, and without scaling the model would be biased toward larger numbers.

The scaler was fit on training data only and then applied to validation and test, to prevent any leakage of information between sets.

7. Conclusion

The preprocessing pipeline successfully cleaned and prepared the dataset for modeling. Missing values were handled, redundant columns were removed, and three useful interaction features were added. The categorical variables were encoded properly, and the data was split and scaled in a way that avoids data leakage. The final dataset has 30 features and is ready to be used for training and evaluating machine learning models.